

SMARTBAND V001

PCB Component & BOM Analysis Report

Band01_Rev1 — Hardware Procurement & Ordering Documentation

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PCB Design Review & Component Procurement

1. Project Overview

This report documents the first task undertaken during the HSHL internship project involving the Smartband V001 (Band01_Rev1) — a wearable health monitoring device. The task predated any software or firmware development and focused entirely on understanding the hardware design, reviewing all PCB components, updating the Bill of Materials (BOM), communicating with PCB/component providers, and placing the manufacturing order.

The Smartband V001 is a compact wrist-worn device designed around the Texas Instruments CC2640R2F Bluetooth Low Energy (BLE) SoC. It integrates multiple biomedical sensors capable of measuring heart rate, blood oxygen saturation (SpO2), temperature, and motion, making it suitable as a research-grade wearable platform aligned with the Multi-sensor Wearable Health Device (MWHD) framework described by Pinheiro et al. (2022).

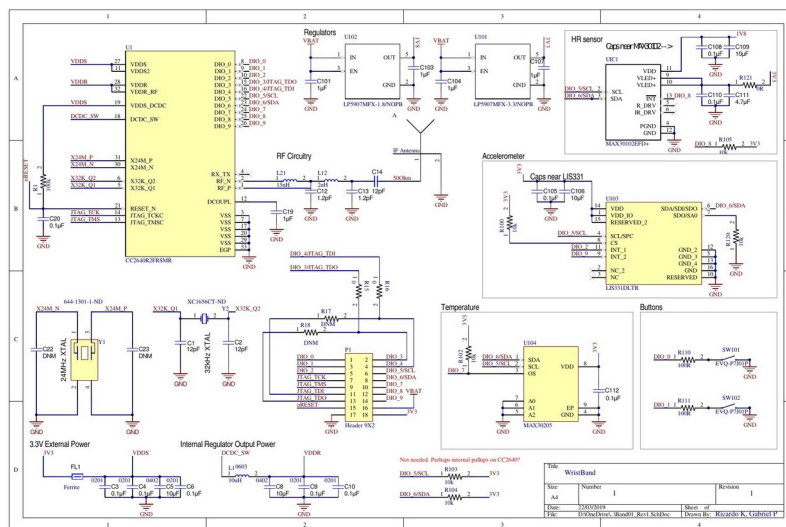
1.1 Design Files Received

The following design package was provided by the supervisor at project inception:

- Band01_Rev1.SchDoc — Altium Designer schematic source file
- Band01_Rev1.PcbDoc — Altium PCB layout source file
- Band01_Rev1.pdf — Exported schematic (colour, A4)
- Band01_Rev1_Mono.pdf — Exported schematic (monochrome)
- Band01_Rev1_3D.pdf — 3D PCB render
- BOM.pdf — Original Bill of Materials
- Fabrication Files/ — Gerber files, drill files for manufacturing

2. PCB Design Overview

The Band01_Rev1 PCB is a compact, two-layer board measuring 34 mm × 40 mm, designed for wrist-band form factor. The layout follows best-practice RF design principles for BLE operation, with a dedicated ground plane and impedance-controlled antenna trace.



2.1 PCB Physical Specifications

Parameter	Value
Board Dimensions	34 mm × 40 mm
Layer Count	2-layer PCB
Via Specification	0.2 mm hole / 0.3 mm copper
Trace Clearance	0.15 mm minimum
Solder Mask Expansion	0.05 mm
RF Antenna Impedance	50 Ω controlled trace
Power Planes	VDDS (22.40 mm wide) + 3V3 (1.16 mm wide)
Antenna Design	PCB trace antenna per TI Application Note AN043 (SWRA117D)
Surface Finish	As per JLCPCB / Aisler specification

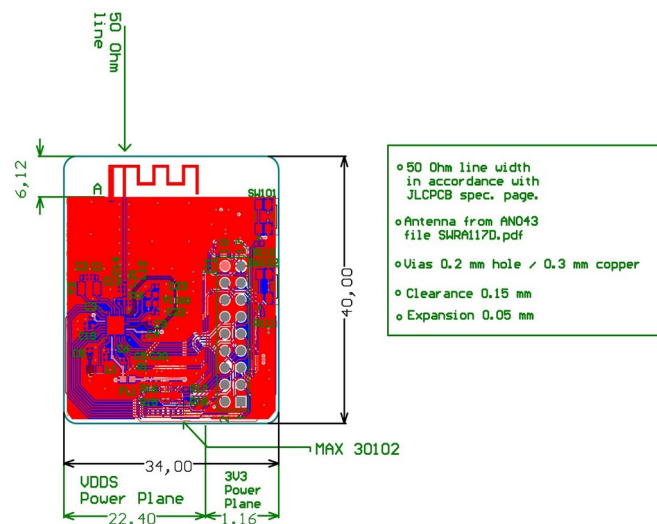


Figure 2: Band01_Rev1 PCB Layout with Power Planes and RF Antenna

2.2 Functional Subsystems

The schematic is organised into the following functional blocks:

- BLE SoC (U1 — CC2640R2FRSMR): Central microcontroller and wireless transceiver
- Power Regulation: Two LP5907MFX LDO regulators (U101: 3.3V, U102: 1.8V)
- RF Circuitry: Impedance-matching network (L21, L12, C12–C14) and PCB antenna
- HR & SpO2 Sensor (UIC1 — MAX30102): Optical heart-rate and blood oxygen sensor
- Accelerometer (U103 — LIS331DLTR): 3-axis MEMS accelerometer via I2C
- Temperature Sensor (U104 — MAX30205): Digital body temperature sensor
- Crystal Oscillators: 24 MHz (Y1) and 32.768 kHz (Y2)
- Debug Header (P1): 9×2 JTAG/SWD programming header
- User Buttons: SW101, SW102 — tactile push buttons

3. Original Bill of Materials

The original BOM was provided as part of the design package (BOM.pdf). It listed 29 line items covering all populated and DNP (Do Not Place) components across 33 individual designators. The document served as the starting point for procurement analysis.

Item	Designator	Comment	Quantity	Footprint
1	A	IF Antenna	1	IF
2	C3, C4, C8, C9, C10	0201 0.1uF	5	0201
3	C19, C101, C103, C104, C107	0402 1uF	5	0402
4	C5, C8, C106, C109	0402 10uF	4	0402
5	C111	0402 4.7uF	1	0402
6	C105, C108, C110, C112	0402 0.1uF	4	0402
7	C12, C13	0201 1.2uF	2	0201
8	C1, C2, C14	0201 12uF	3	0201
9	C20	0201 0.1 F	1	0201
10	C22, C23	DNP	2	0201
11	FL1	BLM18HE152SN1D	1	0603
12	L1	10uH	1	0603
13	L12	2uH	1	L0201
14	L21	15uH	1	L0201
15	P1	Header 8X2	1	HDR2X9
16	R1	100k	1	0402
17	R110, R111	100R	2	0402
18	R100, R102-R105, R120	10k	6	0402
19	R121	0R	1	0402
20	R15, R16	0	2	RES0402
21	R17, R18	DNM	2	RES0402
22	SW101, SW102	EVD-P7J01P	2	SW_EVD-P7J01P_EVD-P7J01P(P55MAY1)
23	U1	CC2640R2FRSMR	1	RSM0032B
24	U101, U102	LP5907MF3.3/NOPB	2	SOT95P280X145-SN
25	U103	LIS33DLTR	1	LGA16SP280X300X300X100
26	U104	MAX30205	1	TDFN3X3
27	UIC1	MAX30102EFD+	1	PSON80P330X560X165-14N
28	Y1	644-1301-1-ND	1	OSC24
29	Y2	XC1656CT-ND	1	32KHZ

Figure 3: Original BOM as received — BOM.pdf from design package

3.1 Original BOM Summary

Category	Components	Count
Capacitors	C1–C14, C19–C23, C101–C112	28
Resistors	R1, R15–R18, R100–R105, R110–R111, R120–R121	14
Inductors / Ferrite	L1, L12, L21, FL1	4
ICs / SoC	U1, U101, U102, U103, U104, UIC1	6
Crystals / Oscillators	Y1, Y2	2
Connectors	P1	1
Switches	SW101, SW102	2
Antenna	A (IF_Antenna)	1 (DNP)
DNP / DNM	C22, C23, R17, R18	4 items not placed

4. BOM Review & Analysis

A detailed review of each component was conducted against the schematic to verify correctness of part numbers, footprints, values, and availability. Key findings from the analysis:

4.1 Components Requiring Attention

Designator	Issue Identified	Action Taken
A (IF Antenna)	Listed but unassigned — PCB trace antenna used instead	Marked DNP in final BOM
C22, C23	Marked DNP — optional bypass capacitors for 24 MHz crystal	Retained as DNP
R17, R18	Marked DNM (Do Not Mount) — JTAG pull resistors not needed	Retained as DNM/DNP
P1 Header	Original BOM listed generic footprint; upgraded to Samtec HTSW-109-07-G-D	Part number updated
U104 (MAX30205)	Supplier listed as generic — sourced from ADI/Maxim (Analog Devices)	Supplier confirmed
UIC1 (MAX30102)	Price volatility flagged — PCBWay noted price subject to change	Price verified at order time
Y1 (24MHz)	Digikey part used in original; switched to muRata equivalent for availability	Substitution applied

5. Updated BOM — PCBWay Quotation

After completing the component review and resolving availability issues, an updated BOM was submitted to PCBWay for assembly quotation (Product No.: T-2Y3W994845A). The quotation covered 3 assembled PCB sets.

5.1 Complete Updated BOM

#	Designator	Value/Part	Qty	Manufacturer	MPN	Unit Price
1	A	IF Antenna	1	—	—	DNP
2	C1, C2	12pF	2	muRata	GRM0335C1H120JA01D	€0.023
3	C3,C4,C6,C9,C10	0.1µF (0201)	5	muRata	GRM033C80J104KE15D	€0.009
4	C5,C8,C106,C109	10µF (0402)	4	muRata	GRM155C80J106ME11D	€0.14
5	C12, C13	1.2pF	2	Kyocera AVX	02013A1R2BAT2A	€0.686
6	C14	12pF	1	muRata	GRM0335C1E120JA01D	€0.067
7	C19,C101,C103,C104,C107	1µF (0402)	5	Samsung	CL05A105MQ5NNNC	€0.005
8	C20	0.1µF (0201)	1	muRata	GRM033C80J104KE15D	€0.037
9	C22, C23	DNP (0201)	0	muRata	GRM0335C1H120JA01D	DNP
10	C105	0.1µF (0402)	1	KEMET	C0402C104M9PACTU	€0.564
11	C108, C110	0.1µF (0402)	2	KEMET	C0402C104M9PACTU	€0.319
1	C111	4.7µF (0402)	1	muRata	GRM155R60J475ME4	€0.063

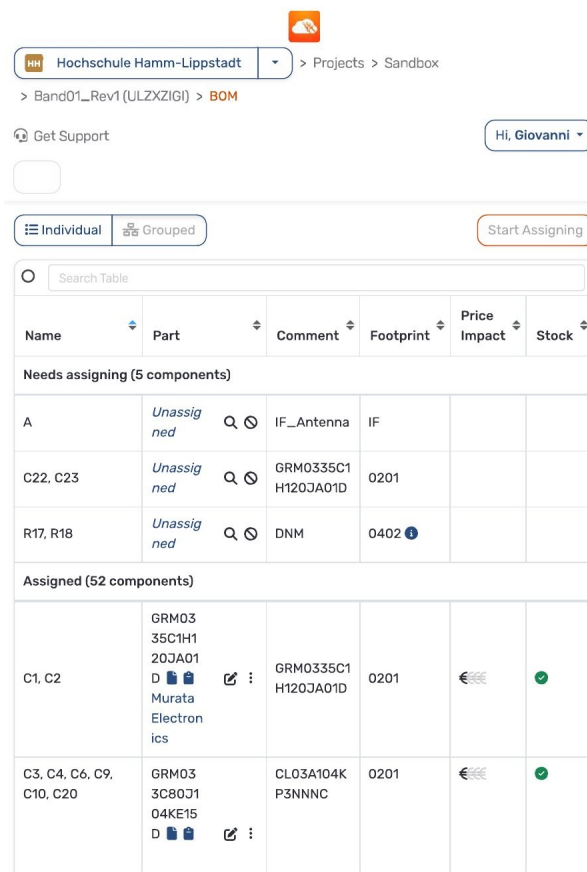
#	Designator	Value/Part	Qty	Manufacturer	MPN	Unit Price
2					7D	
13	C112	0.1µF (0402)	1	KEMET	C0402C104M9PACTU	€0.564
14	FL1	Ferrite Bead	1	muRata	BLM18HE152SN1D	€0.278
15	L1	10µH (0603)	1	Taiyo Yuden	LBMF1608T100K	€0.364
16	L12	2nH (0201)	1	muRata	LQP03TN2N0B02D	€0.036
17	L21	15nH (0201)	1	muRata	LQP03TN15NH02D	€0.077
18	P1	Header 9×2	1	CST	CS-1106-2X9 (L10.0)	€2.214
19	R1	100kΩ	1	Yageo	RC0402JR-10100KL	€0.048
20	R15, R16	0Ω (jumper)	2	Yageo	RC0402JR-070RL	€0.003
21	R17, R18	DNM	0	Yageo	RC0402JR-070RL	DNP
22	R100–R105, R120	10kΩ	6	Yageo	SY0402DD10KP	€0.049
23	R110, R111	100Ω	2	Vishay	CRCW0402100RJNED	€0.20
24	R121	0Ω	1	Yageo	RC0402JR-070RL	€0.006
25	SW101, SW102	Tactile Switch	2	Panasonic	EVQ-P7J01P	€0.512
26	U1	BLE SoC	1	TI	CC2640R2FRSMR	€2.013
27	U101	LDO 3.3V	1	TI	LP5907MFX-3.3/NOPB	€0.255
28	U102	LDO 1.8V	1	TI	LP5907MFX-1.8/NOPB	€0.284
29	U103	Accelerometer	1	ST	LIS331DLTR	€2.013
30	U104	Temp Sensor	1	ADI/Maxim	MAX30205MTA+T	€8.278
31	UIC1	HR/SpO2 Sensor	1	ADI/Maxim	MAX30102EFD+	€15.094
32	Y1	24MHz Crystal	1	muRata	XRCGB24M000F1H17R0	€0.752
33	Y2	32kHz Crystal	1	ECS	ECS-.327-7-34B-TR	€1.361

5.2 PCBWay Cost Breakdown — 3 Sets

Cost Item	Amount (USD)
Component Cost	\$116.33
Assembly Cost	\$88.00
PCB Fabrication Cost	\$25.97
Discount (V0 Member)	\$0.00
Total — 3 Sets	\$230.30
Unit Cost per Board	≈ \$76.77
Lead Time	26–28 working days

6. Aisler BOM — Final Assembled PCB

In parallel with the PCBWay process, the project was also registered on the Aisler platform (Hochschule Hamm-Lippstadt workspace) under project Band01_Rev1 (ULZXZIGI). Aisler provides an integrated component assignment workflow with real-time stock and pricing from supported distributors. The Aisler BOM represents the final component assignment state used for the assembled PCB order.



Name	Part	Comment	Footprint	Price Impact	Stock
Needs assigning (5 components)					
A	Unassigned	IF_Antenna	IF		
C22, C23	Unassigned	GRM0335C1 H120JA01D	0201		
R17, R18	Unassigned	DNM	0402		
Assigned (52 components)					
C1, C2	GRM03 35C1H1 20JA01 D Murata Electronics	GRM0335C1 H120JA01D	0201	€	✓
C3, C4, C6, C9, C10, C20	GRM03 3C80J1 04KE15 D	CL03A104K P3NNNC	0201	€	✓

Figure 4: Aisler BOM — Component assignment view, Page 1

Name	Part	Comment	Footprint	Price Impact	Stock
	Murata Electronics				
C5, C8, C106, C109	GRM155C80J106ME1D Murata Electronics	GRM155R61A105KE15D	0402	€	✓
C12, C13	GRM0335C1E1R2BA01D Murata Electronics	CL03A104K P3NNNC	0201	€	✓
C14	GRM0335C1E120JA01D Murata Electronics	CL03A104K P3NNNC	0201	€	✓
C19, C101, C103, C104, C107	CL05A105MQ5NNNC Samsung	GRM155R61A105KE15D	0402	€	✓
C105, C108, C110, C112	C0402C104M9PACTU KEMET	GRM155R61A105KE15D	0402	€	✓

Figure 5: Aisler BOM — Component assignment view, Page 2

6.1 Aisler Component Status

Status	Count	Notes
Assigned (in stock)	52 components	All sourced from supported distributors
Needs Assigning	5 components	A (DNP), C22/C23, R17/R18
DNP (Do Not Place)	4 items	Antenna, optional caps, unused resistors
Total Unique Items	57 items	Including DNP entries

6.2 Aisler Pricing Tiers

Quantity	Total Price	Unit Price / Board
3 sets	€376.69	€125.56
5 sets	€439.65	€87.93
10 sets	€590.66	€59.07
50 sets	€1,893.43	€37.87
100 sets	€3,249.84	€32.50
200 sets	€6,183.19	€30.92
500 sets	€14,134.49	€28.27

Note: Prices are best-price net-prices from supported distributors including sourcing expenses and handling fees. Valid for 24 hours from quotation.

7. Key Component Highlights

The following section highlights the most technically significant components on the board and their roles in the wearable health monitoring application.

Component	Part Number	Function	Interface
CC2640R2FRSMR (U1)	TI CC2640R2F	BLE 5.0 SoC — ARM Cortex-M3 @ 48 MHz, 128KB Flash, 20KB RAM	JTAG / SWD
MAX30102EFD+ (UIC1)	Maxim MAX30102	Heart Rate & SpO2 optical sensor — integrated LED driver, 18-bit ADC	I2C
LIS331DLTR (U103)	ST LIS331D	3-axis $\pm 24g$ accelerometer — motion/activity detection	I2C
MAX30205MTA+T (U104)	Maxim MAX30205	$\pm 0.1^{\circ}\text{C}$ accurate body temperature sensor	I2C
LP5907MFX-3.3 (U101)	TI LP5907	Ultra-low noise 3.3V LDO regulator — powers CC2640R2F and peripherals	—
LP5907MFX-1.8 (U102)	TI LP5907	Ultra-low noise 1.8V LDO regulator — powers HR sensor	—

8. Procurement Process & Provider Communication

8.1 Process Summary

The procurement process followed these sequential steps:

- Step 1: Received design ZIP package from supervisor (giovannix) via private GitHub repository
- Step 2: Reviewed schematic (Band01_Rev1.pdf) and extracted all component information
- Step 3: Cross-referenced original BOM.pdf against schematic to identify gaps and errors
- Step 4: Researched component availability on Mouser, DigiKey, LCSC, and Aisler
- Step 5: Resolved 5 components requiring action (see Section 4.1)
- Step 6: Prepared updated BOM in PCBWay format for assembly quotation
- Step 7: Submitted BOM to PCBWay — received quote for 3 sets at \$230.30 total
- Step 8: Registered project on Aisler platform under HSHL workspace
- Step 9: Assigned all available components in Aisler (52/57 assigned)
- Step 10: Confirmed order placement for PCB fabrication and assembly

8.2 Provider Summary

Provider	Role	Platform	Outcome
PCBWay	PCB fabrication + assembly	pcbway.com	Quotation received — \$230.30 / 3 sets, 26–28 day lead time
Aisler	PCB fabrication + assembly (EU)	aisler.net	Component assignment complete, €376.69 / 3 sets

Provider	Role	Platform	Outcome
Mouser Electronics	Component sourcing	mouser.com	Used for availability cross-check
DigiKey	Component sourcing	digikey.com	Y1 crystal original part source
LCSC	Component sourcing	lcsc.com	Used for KEMET capacitor pricing

9. Conclusion & Next Steps

This phase of the project successfully delivered a complete hardware understanding of the Smartband V001 PCB, a corrected and validated Bill of Materials, provider communications, and a confirmed manufacturing order through both PCBWay and Aisler platforms.

The board was ordered in quantities of 3 sets to allow for prototyping, testing, and potential rework. Upon delivery, the hardware will be prepared for firmware flashing using Texas Instruments Code Composer Studio (CCS) targeting the CC2640R2F BLE SoC.

Immediate Next Steps

- Await PCB delivery from PCBWay / Aisler (26–28 day lead time)
 - Visual and electrical inspection of assembled boards upon receipt
 - Set up Code Composer Studio (CCS) environment for CC2640R2F
 - Flash and validate firmware build — proceed to Task 2 (Firmware Analysis)
 - Begin sensor integration and BLE data transmission testing
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